

Revised Load Interconnection System Impact Study Report

For Queue Project Q17-0845

Quantico 138 kV
December 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0845. The Interconnection Customer proposes a 133.6 MW large load (data center) facility interconnecting at the Quantico 138 kV substation in the Virginia area. The request was received on August 05, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 345 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,656,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Westwood Generation Llc to Morganville 345kV L
Contingency: Wigwam to Upsco 345kV line
Loading: 106.5% (Rating: 1,222 MVA)
DFAX: 5.1%
Pre-project loading: 105.9%
2. Facility: Clark County to Artesia Country Club 345kV
Contingency: Grimmway-Malaga to Sleepy Hollow 138kV
Loading: 105.6% (Rating: 1,291 MVA)
DFAX: 9.5%
Pre-project loading: 104.6%
3. Facility: Novec Country Club Substation to Lindendale D.P 138kV
Contingency: Possum Pt 345/138kV Auto
Loading: 103.1% (Rating: 221 MVA)
DFAX: 43.5%
Pre-project loading: 76.8%
4. Facility: Richland 500/230kV Transformer
Contingency: Richland to Troy Energy Llc 500kV Line
Loading: 103.0% (Rating: 1,398 MVA)
DFAX: 6.0%
Pre-project loading: 102.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Westwood Generation Llc to Morganville 345kV L	Reconductor line	\$11,792,000	1% (\$160,000)
Artesia Countryclub to Clark County 345kV	Reconductor line	\$21,153,000	1% (\$276,000)
Novec Country Club Substation to Lindendale D.P 138kV	Reconductor line	\$2,077,000	100% (\$2,077,000)
Richland 500/230kV Transformer	Replace transformer with higher-rated unit	\$18,390,000	1% (\$143,000)
	Total allocated to project		\$2,656,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.